

SOLUTION DESIGN DOCUMENT (SDD) v2.0

AI vs Human Content Detection System

Project Name: AI vs Human Content Detection System

Document Type: Solution Design Document (SDD)

Version: 2.0 Final

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Status: Approved for Client Handover

Date: June 2026

DOCUMENT CONTROL

Version	Date	Author	Description
1.0		Project Initiation Data Science Team	Initial Solution Design
2.0	June 2026	M. Sagar Reddy	Final Implemented Architecture

APPROVAL MATRIX

Role	Responsibility	Status
Client Stakeholder	Solution Approval	Pending
Data Science Lead	Technical Approval	Completed
Deployment Team	Deployment Review	Client Owned

1. PURPOSE

The purpose of this document is to describe the final technical architecture, machine learning pipeline, feature engineering design, validation framework, serialization architecture, and deployment packaging strategy implemented for the AI vs Human Content Detection System.

This document reflects the final solution delivered to the client rather than the originally proposed architecture.

2. SOLUTION OBJECTIVES

The solution was designed to:

Business Objectives

- Detect AI-generated content
- Protect human-authored content
- Support moderation workflows
- Enable scalable content verification

Technical Objectives

- Prevent data leakage
 - Improve generalization
 - Support deployment packaging
 - Provide low-latency inference
 - Support future deployment integration
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3. FINAL SOLUTION ARCHITECTURE

End-to-End Architecture

Raw Text Input

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Input Validation

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Text Cleaning

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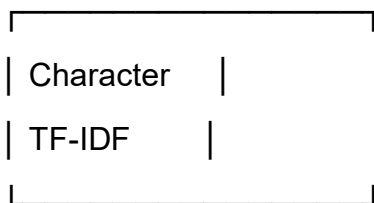
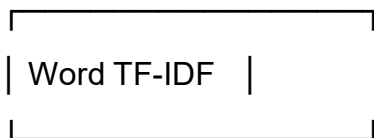


Feature Engineering

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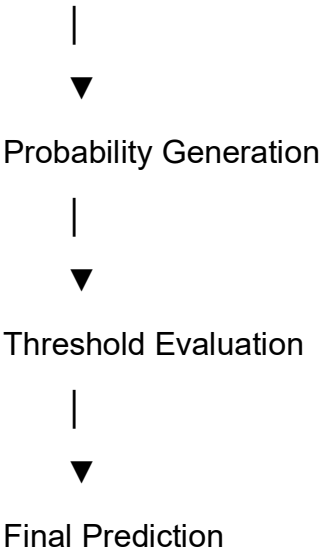
Feature Union



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Logistic Regression



4. DATA ARCHITECTURE

Source Dataset

Attribute	Value
Dataset Type	Structured CSV
Total Records	2,000
Domains	Academic, News, Social
Target Classes	AI, Human

Input Features

Original Dataset Features

- text_content
- word_count
- avg_sentence_length
- domain
- topic_hint
- source_model
- generation_method

Target Variable

- label
- Values:

AI

Human

5. DATA LEAKAGE ASSESSMENT

One of the most critical phases of the project involved identifying and eliminating data leakage.

Leakage Risks Identified

source_model

Problem:

Directly revealed content origin.

Decision:

Excluded.

generation_method

Problem:

Unavailable during inference.

Decision:

Excluded.

topic_hint

Problem:

Not guaranteed during production prediction.

Decision:

Excluded.

domain

Problem:

Not always available during deployment.

Decision:

Excluded from production pipeline.

6. DATA SPLITTING STRATEGY

Initial Approach

Original split:

`train_test_split()`

Result:

Unexpectedly perfect performance.

Investigation Findings

The project identified potential content duplication patterns causing inflated performance.

Final Solution

`GroupShuffleSplit()`

Grouping Variable:

`text_content`

Purpose:

Prevent duplicate content appearing in both train and test datasets.

Final Dataset Split

Dataset Records

Training 1,592

Testing 408

7. TEXT PREPROCESSING DESIGN

The preprocessing layer standardizes text before feature extraction.

Processing Pipeline

Step 1

Convert text to lowercase.

Example:

THIS STUDY INVESTIGATES AI

↓

this study investigates ai

Step 2

Remove punctuation.

Step 3

Remove special characters.

Step 4

Normalize whitespace.

Output

clean_text

8. FEATURE ENGINEERING DESIGN

The final production pipeline utilizes a purely NLP-based architecture.

Word-Level TF-IDF

Purpose:

Capture vocabulary usage and phrase patterns.

Configuration:

```
TfidfVectorizer(  
    stop_words='english',  
    ngram_range=(1,2),  
    max_features=5000,  
    sublinear_tf=True  
)
```

Generated Features

- Unigrams
- Bigrams

Examples:

climate

research

future research
machine learning

Character-Level TF-IDF

Purpose:

Capture writing style and token construction patterns.

Configuration:

```
TfidfVectorizer(  
    analyzer='char_wb',  
    ngram_range=(3,5),  
    max_features=10000,  
    sublinear_tf=True  
)
```

Examples

tion
ing
ated
study

9. FEATURE UNION DESIGN

The final architecture combines both feature spaces.

```
FeatureUnion([  
    ("word_tfidf"),  
    ("char_tfidf")  
])
```

Benefits:

- Vocabulary patterns
 - Writing style patterns
 - Improved generalization
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10. MODEL DEVELOPMENT STRATEGY

Multiple candidate models were evaluated during development.

Candidate Models

Logistic Regression

Advantages:

- Fast
 - Interpretable
 - Strong TF-IDF performance
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Random Forest

Advantages:

- Non-linear modeling

Challenges:

- Lower generalization
-

XGBoost

Advantages:

- High internal accuracy

Challenges:

- Generalization concerns
-

Final Model Selection

Selected:

Word + Character TF-IDF Logistic Regression

Reason:

Best external validation performance.

11. FINAL MODEL CONFIGURATION

LogisticRegression(
 max_iter=3000,
 class_weight='balanced',
 C=1.0

)

12. THRESHOLD OPTIMIZATION

Default threshold:

0.50

Final threshold:

0.44

Purpose:

Improve AI detection performance on external validation data.

Decision Logic

if ai_probability >= 0.44:

 prediction = "AI"

else:

 prediction = "Human"

13. VALIDATION FRAMEWORK

Internal Validation

Training/Test Evaluation

Results:

Metric	Score
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Accuracy	100%
----------	------

Precision	100%
-----------	------

Recall	100%
--------	------

F1 Score	100%
----------	------

External Validation

Independent dataset:

Metric	Score
--------	-------

Accuracy	82.05%
----------	--------

Precision	81.82%
-----------	--------

Metric	Score
--------	-------

Recall	91.67%
--------	--------

F1 Score	86.47%
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Selection Principle

External validation performance was prioritized over internal metrics.

14. SERIALIZATION ARCHITECTURE

The final deployment package consists of three serialized artifacts.

Artifact 1

final_ai_detector.pkl

Contains:

- Word TF-IDF
 - Character TF-IDF
 - Feature Union
 - Logistic Regression
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Artifact 2

decision_threshold.pkl

Contains:

0.44

Artifact 3

model_metadata.pkl

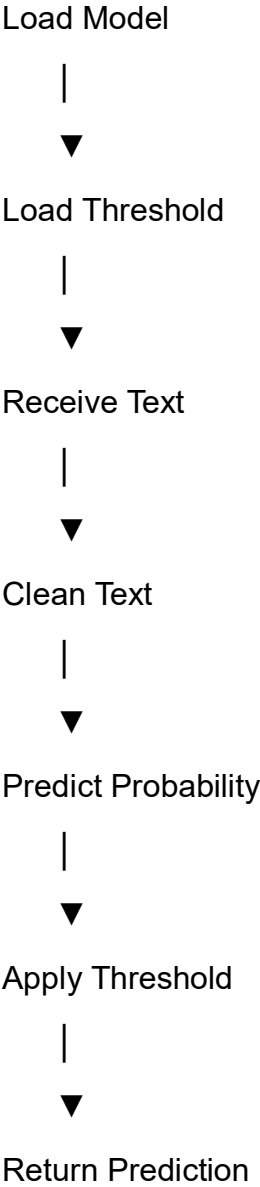
Contains:

- Accuracy
- Precision
- Recall
- F1 Score
- Threshold
- Sample Counts

15. DEPLOYMENT ARCHITECTURE

Client deployment team owns production deployment.

Deployment Workflow



16. TECHNICAL RISKS

Risk	Impact
AI Model Evolution	Reduced Accuracy
Domain Drift	Generalization Issues
New Writing Styles	Performance Degradation

17. FUTURE ENHANCEMENTS

Recommended improvements:

- Larger training dataset
 - Continuous retraining
 - Additional external validation
 - Multilingual support
 - API deployment layer
 - Monitoring framework
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18. PROJECT CLOSURE STATEMENT

The final architecture successfully delivered a production-ready AI vs Human content detection solution incorporating leakage prevention, hybrid TF-IDF feature engineering, threshold optimization, external validation, and deployment packaging.

The solution has been serialized, validated, and approved for client handover.

APPENDIX A – FINAL ARCHITECTURE DIAGRAM

Text Input

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Cleaning

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Feature Union

└─ Word TF-IDF

└─ Character TF-IDF

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Logistic Regression

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Threshold 0.44

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Prediction
